# THE CHINESE UNIVERSITY OF HONG KONG
# ECLT5930B & SEEM5740B: Engineering Economics
# Final Examination 2025

**Instructions:**

- *This examination consists of **3 Sections**. Answer all questions. The total mark is **100**. The duration of the examination is **120 minutes**.*
- *Answers must be given in English. Answers without sufficient intermediate steps may lose some marks.*
- *Unless otherwise specified, numerical answers should be either **exact** or **corrected to 3 significant figures**.*
- *The exam paper and draft worksheets must be returned along with answer book at the end of the exam.*

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## Section 1: Choice Questions (15 marks). Choose one appropriate answer in each of the five questions.

1. (3 marks) Identify a wrong statement (D)

   (A) Depreciation usually refers to cost recovery in calculation of taxable incomes.

   (B) Depreciation is book cost which is purely for taxation purpose and does not involve actual cash flow.

   (C) The time and amount of depreciation/cost recovery are usually determined by a company which maximizes relief of income taxes in the duration of a project.

   (D) MARR should be considered in calculation of depreciation/cost recovery over a specified time period.

2. (3 marks) If the present worth of a project is a convex function of an uncertainty parameter, then using the expected value of the parameter to calculate the present worth may (B)

   (A) overestimate the expected present worth.

   (B) underestimate the expected present worth.

   (C) precisely determine the expected present worth.

   (D) be difficult to say.

3. (3 marks) Identify a wrong statement (D)

   (A) Real interest rate cannot be used to calculate taxable incomes.

   (B) Real interest rate is usually not used for calculating cash flows in accounting.

   (C) Real interest rate is important because it is concerned with actual dollars.

   (D) Present worth of a project based on real interest rate is usually lower than the one calculated with market interest rate.

4. (3 marks) Identify a wrong statement (C)

   (A) Expected monetary value does not necessarily capture a decision maker's (DM's) risk preferences unless the DM is risk neutral.

   (B) The first/second order stochastic dominance relationship between two products is purely determined by the stochastic natures of the products instead of customer's preferences.

   (C) Consider present worth of two projects A and B. If PW(A) is greater or equal to PW(B) in every scenario, then the cumulative distribution function (CDF) of PW(A) is greater or equal to the CDF of PW(B).

   (D) In (C), project A dominates project B.

5. (3 marks) Identify a wrong statement (C)

   In decision making problems,

   (A) Decision makers are usually risk averse on gains.

   (B) Decision makers are usually risk taking on losses.

   (C) Decision maker's behaviour can always be described by von Neumann-Morgenstern's expected utility theory.

   (D) Decision maker's behaviour can be described by von Neumann-Morgenstern's expected utility theory only if the DM's preferences satisfy four specified axiomatic properties.

## Section 2: Short Explanation Questions (25 marks)

1. (9 marks) A project has three uncertain factors: useful life, annual revenue, and annual expense.

   (a) (5 marks) Explain the economic meaning of the curves in Figure 1.

   (b) (4 marks) Suppose that each factor has 3 scenarios, and the annual worth of the project in each combination of the scenarios of the factors are listed in Figure 2. Explain the respective scenarios under which the circled numbers 3900 and 30650 are calculated.

   **Solution:**

   (a)

   Center Point (0%): Shows the project's expected profitability (Annual Worth $\approx 5,000$) when all factors are at their most likely values.

   X-Axis: Shows the percentage change (+/-) of a single factor from its most likely value.

   Y-Axis: Shows the resulting Annual Worth (profitability) in thousands of dollars.

   Each Line: Tracks how Annual Worth changes when only that specific factor changes.

   Steepness: Steeper lines (like Revenue and Expenses) indicate factors the project's profitability is very sensitive to; flatter lines (like Useful Life and MARR) indicate factors it's less sensitive to.

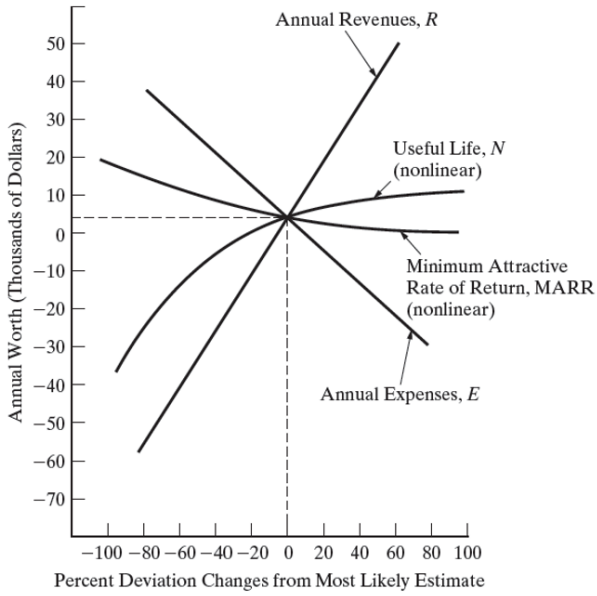


Figure 1: Section 2, Question 1(a)

In summary, the project's profitability is highly dependent on **Annual Revenues** and **Annual Expenses** – small changes in these have a large impact. Profitability is much less sensitive to changes in the project's **Useful Life** and the **Minimum Attractive Rate of Return (MARR)**.

(b) 3900:

useful life: Worst;

annual revenue: Worst;

annual expense: Best.

30650:

useful life: M-L;

annual revenue: Best;

annual expense: Worst.

| Annual Revenue | Annual Expense | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | Best Scenario | | | Most-likely | | | Worst Scenario | | |
| | Useful Life | | | Useful Life | | | Useful Life | | |
| | Best | M-L | Worst | Best | M-L | Worst | Best | M-L | Worst |
| Best | 73995 | 67650 | 63900 | 50995 | 44650 | 40900 | 36995 | 30650 | 26900 |
| M-L | 34000 | 27650 | 23900 | 10995 | 4650 | 900 | -3005 | -9350 | -13100 |
| Worst | 14000 | 7650 | 3900 | -9005 | -15350 | -19100 | -23005 | -29350 | -33100 |

Figure 2: Section 2, Question 1(b)

2. (5 marks) Use $(R\$)_k = (R\$)_b \times (F/P, i_r\%, k-b)$ and $(A\$)_k = (R\$)_b \times (F/P, i_m\%, k-b)$ to derive the equation $(R\$)_k = (A\$)_k \times (P/F, f\%, k-b)$.

**Solution**:

$$\frac{(R\$)_k}{(A\$)_k} = \frac{(R\$)_k}{(R\$)_b} / \frac{(A\$)_k}{(R\$)_b} = \frac{(F/P, i_r\%, k-b)}{(F/P, i_m\%, k-b)} = (P/F, f\%, k-b).$$

3. (6 marks) Consider a decision making problem where the decision maker needs to decide whether to adopt a new design or not. Adopting a new design may lead to four service levels with specified probabilities and outcomes of PW detailed in Table 1. Assume that a test can be made and the test will tell the DM for sure the outcome of the service level. Explain the decision process (path and nodes) shown in Figure 3.

| Service level | Probability | Annual saving | PW |
|---|---|---|---|
| 1 | 0.25 | 3,470 | 3,538 |
| 2 | 0.40 | 2,920 | 1,614 |
| 3 | 0.25 | 2,310 | -520 |
| 4 | 0.10 | 1,560 | -3,143 |

Table 1: Section 2, Question 3

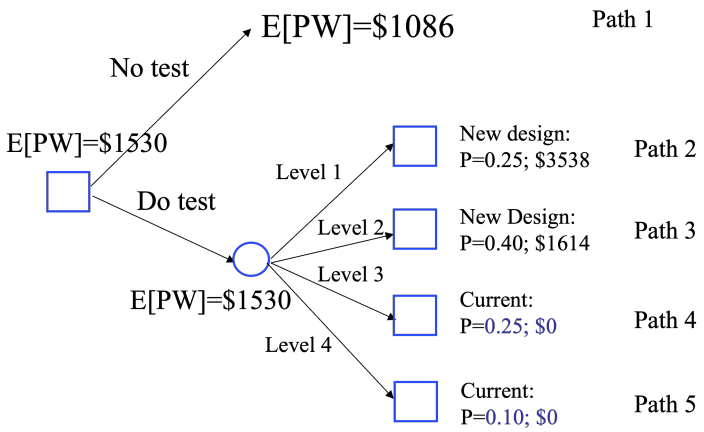


Figure 3: Section 2, Question 3

**Solution.**

- **Left Square** (Initial Decision Node): The process starts here. The decision maker (DM) must choose an **initial (first) action**, between 'No Test' (upper path) and 'Do Test' (lower path).
- **Path 1**: represents the decision to not perform the test and make the choice to adopt the new design. The expected of PW is calculated based only on the probabilities and PW in each service level. The decision is made under uncertainty in service level.
- **Circle** (Chance Node): If the DM chooses 'Do Test' (lower path), they reach this node. It represents the uncertainty of the test result. Nature determines which service level (1, 2, 3, or 4) the test will reveal, according to the probabilities in the table.

- **Right Squares** (Subsequent Decision Nodes): Each branch from the circle (representing a specific test outcome/service level) leads to one of these decision nodes. Here, after knowing the test result, the DM makes **a second decision** to choose 'new design' or 'current': (a) if $PW > 0$ in corresponding service levels, then adopting 'New Design' (resulting in the PW for that specific service level in **path 2 and path 3**); (b) if $PW < 0$ in corresponding service levels, then choosing 'current' ('don't adopt'), resulting in \$0 PW, as **path 4 and path 5**).

Essentially, the figure illustrates the path: Step 1: Initial Decision (Test?) $\rightarrow$ Outcome 1 (if do test): Test Outcome (service level) $\rightarrow$ Step 2: Second Decision (Adopt new design?) $\rightarrow$ Outcome 2: Final Outcome (PW).

4. (5 marks) Explain the difference between the Internal Rate of Return (IRR) and the External Rate of Return (ERR) in terms of their definitions and underlying assumptions. When might ERR provide a more realistic measure of a project's profitability than IRR?

   **Solution**: The Internal Rate of Return (IRR) is the interest rate that makes the net present value (NPV) of all cash flows from a project equal to zero. It assumes that all intermediate cash flows are reinvested at the IRR itself, which may not be realistic if the IRR is high.

   The External Rate of Return (ERR), on the other hand, is a modified rate that assumes intermediate cash flows are reinvested at a known external rate, such as the firm's cost of capital or minimum acceptable rate of return (MARR). All net cash outflows are converted to PW and all net cash inflows are converted to FW. The ERR is the interest rate under which the above two quantities are equivalent.

   The Calculation of IRR is difficult. The solution may not be unique. There may not be feasible solutions.

   ERR often provides a more realistic evaluation when reinvestment opportunities are limited or when the IRR is unrealistically high, because it reflects actual market or company reinvestment conditions.

## Section 3: Calculation Questions (60 marks)

1. (14 marks) If the interest rate is 10% per year, calculate the expected present worth of introducing new product and what decision would you make based on the decision tree diagram in Figure 4?

   **Solution.** The present worth of introducing new product is:

$$\begin{aligned} PW &= -1000000(P/F, 10\%, 2) + 0.6 * 200000 * (P/A, 10\%, 6)(P/F, 10\%, 2) \\ &\quad +0.4 * 280000 * (P/A, 10\%, 6)(P/F, 10\%, 2) \\ &= -826446.28 + 0.6 * 871052.14/1.1^2 + 0.4 * 1219473.00/1.1^2 = 8,611.97. \end{aligned}$$

   Therefore, we should introduce new product.

2. (11 marks) With reference to the data provided in Table 2, recommend the preferred alternative by using

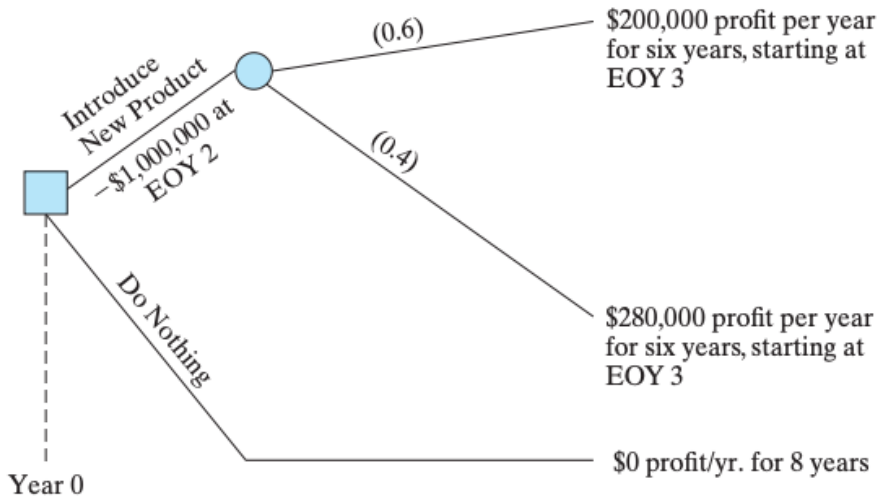


Figure 4: Question 3

| Attribute | Alternative | | | Worst Acceptable Value |
|---|---|---|---|---|
| | Vendor I | Vendor II | Vendor III | |
| A. Reduction in throughput time | 75% | 70% | 84% | 50% |
| B. Flexibility | Good | Excellent | Excellent | Good |
| C. Reliability | Excellent | Good | Good | Good |
| D. Quality | Good | Excellent | Excellent | Excellent |
| E. Cost of system (PW of life - cycle cost) | $270,000 | $310,000 | $280,000 | $350,000 |

Table 2: Section 3, Question 2

(a) (4 marks) dominance,

**Solution.** Vendor II is dominant by Vender III. Recommend Vendor I and Vendor III.

(b) (3 marks) satisficing,

**Solution.** Vendor I is eliminated for the attribute of quality. Recommend Vendor II and Vendor III.

(c) (4 marks) non-dimensional scaling, with weights $w_A = 0.15, w_B = 0.2, w_C = 0.3, w_D = 0.2, w_E = 0.15$.

Weighted sum of Vendor I is:

$$0.15 * 0.5 + 0.2 * 0 + 0.3 * 1 + 0.2 * 0 + 0.15 * 1 = 0.525,$$

Weighted sum of Vendor II is:

$$0.15 * 0 + 0.2 * 1 + 0.3 * 0 + 0.2 * 1 + 0.15 * 0 = 0.4,$$

| Attribute | Alternative | | | Worst Acceptable Value |
|---|---|---|---|---|
| | Vendor I | Vendor II | Vendor III | |
| A. Reduction in throughput time | 0.5 | 0 | 1 | 50% |
| B. Flexibility | 0 | 1 | 1 | Good |
| C. Reliability | 1 | 0 | 0 | Good |
| D. Quality | 0 | 1 | 1 | Excellent |
| E. Cost of system (PW of life - cycle cost) | 1 | 0 | 0.5 | \$350,000 |

Table 3: Section 3, Question 2, Answer

Weighted sum of Vendor III is:

$$0.15 * 1 + 0.2 * 1 + 0.3 * 0 + 0.2 * 1 + 0.15 * 0.5 = 0.625,$$

Recommend Vendor III.

3. (15 marks) You are faced with a decision as to whether to purchase an insurance product on a hire car. The probabilities of different amounts of damage are as follows

| Amount of damage in dollars | 500 | 1000 | 2000 | 4000 | 8000 |
|---|---|---|---|---|---|
| Probability | 0.03 | 0.03 | 0.02 | 0.01 | 0.01 |

(a) (6 marks) Assume that you have a linear utility function and so are concerned to maximize expected profit (or minimize expected loss). What is the largest amount you should pay for insurance cover given that there is an excess of \$500 (i.e. if there is damage of an amount $X > 500$ then you pay \$500 and the insurance company pays $X - 500$)?

Answers:

The expected loss given an excess of \$500 is

$$0.03 \times 500 + 0.02 \times 1500 + 0.01 \times 3500 + 0.01 \times 7500 = 15 + 30 + 35 + 75 = \$155.$$

So the largest pay for the insurance is \$155 **[6 marks]**

(b) (9 marks) How does your answer to (i) change if utilities are calculated as follows: a cost of $y$ gives a reduction in utility of $\left(\frac{y}{1000}\right)^2$.

Answers:

We equate the loss of utility in paying an amount $y$ for the insurance with the expected loss of utility given by

$$0.03 \times 0.5^2 + 0.02 \times 1.5^2 + .01 \times 3.5^2 + 0.01 \times 7.5^2 = 0.7375.$$

**[6 marks]**

Therefore

$$(y/1000)^2 = 0.7375$$

and $y = \$859$. **[3 marks]**

There is another way to answer this question.

Without paying the insurance, the expected loss of utility is

$$0.03 \times 0.5^2 + 0.03 \times 1^2 + 0.02 \times 2^2 + 0.01 \times 4^2 + 0.01 \times 8^2 = 0.9175.$$

By paying the insurance, the expected loss is

$$0.1 \times (0.5 + y/1000)^2 + 0.9 \times (y/1000)^2.$$

**[6 marks]**

Equalizing the two quantities, we arrive at

$$z^2 + 0.2z = 0.8925$$

where $z := y/1000$. Solving the equation, we obtain $z = 0.85$ and hence $y = \$859$. **[3 marks]**

4. (20 marks) A company is considering to install a new equipment which costs \$100,000 with useful life of 3 years. Generating annual revenue in real dollars \$45,000 per year . The market value of the equipment at the end of the study period is 30,000 in real dollars. Assume the effective tax rate is 30%; Average inflation rate: 5%; Annual maintenance cost for the first 3 years is 5,000 in actual dollars. The equipment can be depreciated for two years as 50000 and 50000 in actual dollars.

   (a) (9 marks) Calculate before tax cash flows and after tax cash flows.

   (b) (3 marks) Based on the calculations in (a), develop a cash flow diagram.

   (c) (2 marks) Suppose that MARR in actual dollars is 21.5%. Calculate the MARR in real dollars.

   (d) (6 marks) Suppose that MARR in actual dollars is 21.5%. Calculate the present worth of investment in real dollars.

**Solution.**

(a) The table for BTCFs and ATCFs is as follows:

| year | CF | Cost | BTCF | $d_k$ | TI | -tTI | ATCF |
|---|---|---|---|---|---|---|---|
| 0 | -100000 | 0 | -100000 | | | | -100000 |
| 1 | 47250 | -5000 | 42250 | 50000 | -7750 | 2325 | 44575 |
| 2 | 49612.5 | -5000 | 44612.5 | 50000 | -5387.5 | 1616.25 | 46228.75 |
| 3 | 52093.125 | -5000 | 47093.125 | | 47093.125 | -14127.9375 | 32965.1875 |
| 3 | 34728.75 | | 34728.75 | | 34728.75 | -10418.625 | 24310.125 |

where $CF$ means cash flow, and $CF_k = 45000 * 1.05^k$ for $k = 1, 2, 3$, $CF_3 = 30000 * 1.05^3$, cost = maintenance cos = 5000, BTCF = CF + cost, $d_k$ means depreciation, TI means taxable income and $TI = BTCF - d_k$, and $tTI = t(BTCF - d_k)$, ATCF= BTCF + tTI.

(b) Year 0: $-100,000$. Year 1-3: Revenue $45000 \times 1.05^k$ for $k = 1, 2, 3$, maintenance: -5000. Year 3: The actual cash flow of salvage value is calculated as:

$$SV_A = SV_R * (1+f)^3 = 30000 * (1+0.05)^3 = 34728.75.$$

The cash flow diagram is

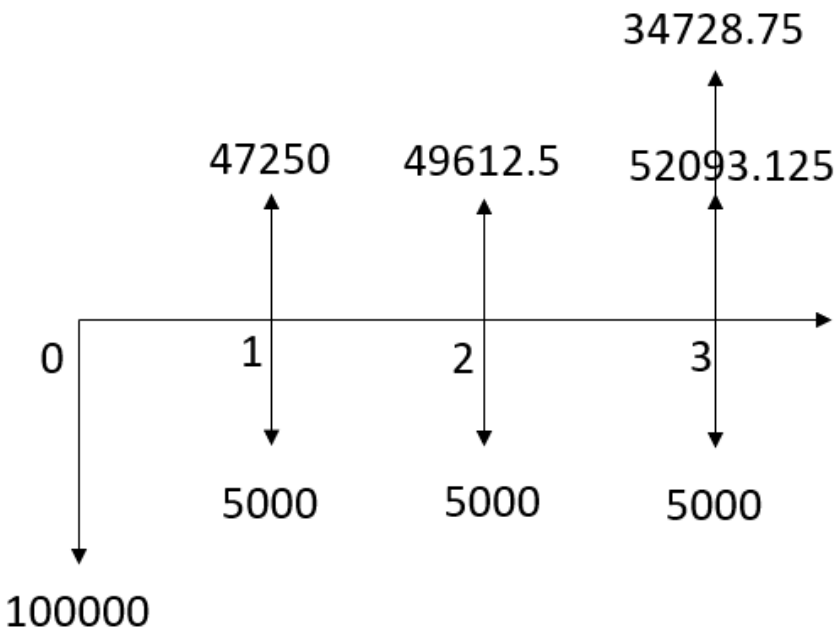


Figure 5: Q-4-2

(c) $MARR_r = \frac{MARR_A - f}{1+f} = \frac{0.215-0.05}{1+0.05} = 0.15714.$

(d) Solution 1: First, convert the ATCF (A) to ATCF (R):
Year 1: $44575/1.05 = 42452.38$;
Year 2: $\frac{46228.75}{1.05^2} = 41930.84$;
Year 3: $\frac{32965.1875+24310.125}{1.05^3} = 49476.56841$.

$$\begin{aligned} PW &= 42452.38/(1+0.15714)^1 + 41930.84/(1+0.15714)^2 \\ &\quad +49476.56841/(1+0.15714)^3 \\ &= 99936.16. \end{aligned}$$

Solution 2:

$$\begin{aligned} PW_R &= PW_A \\ &= 44575/(1+0.215)^1 + 46228.75/(1+0.215)^2 \\ &\quad +(32965.1875 + 24310.125)/(1+0.215)^3 \\ &= 99935.68047. \end{aligned}$$

**— End of Paper —**